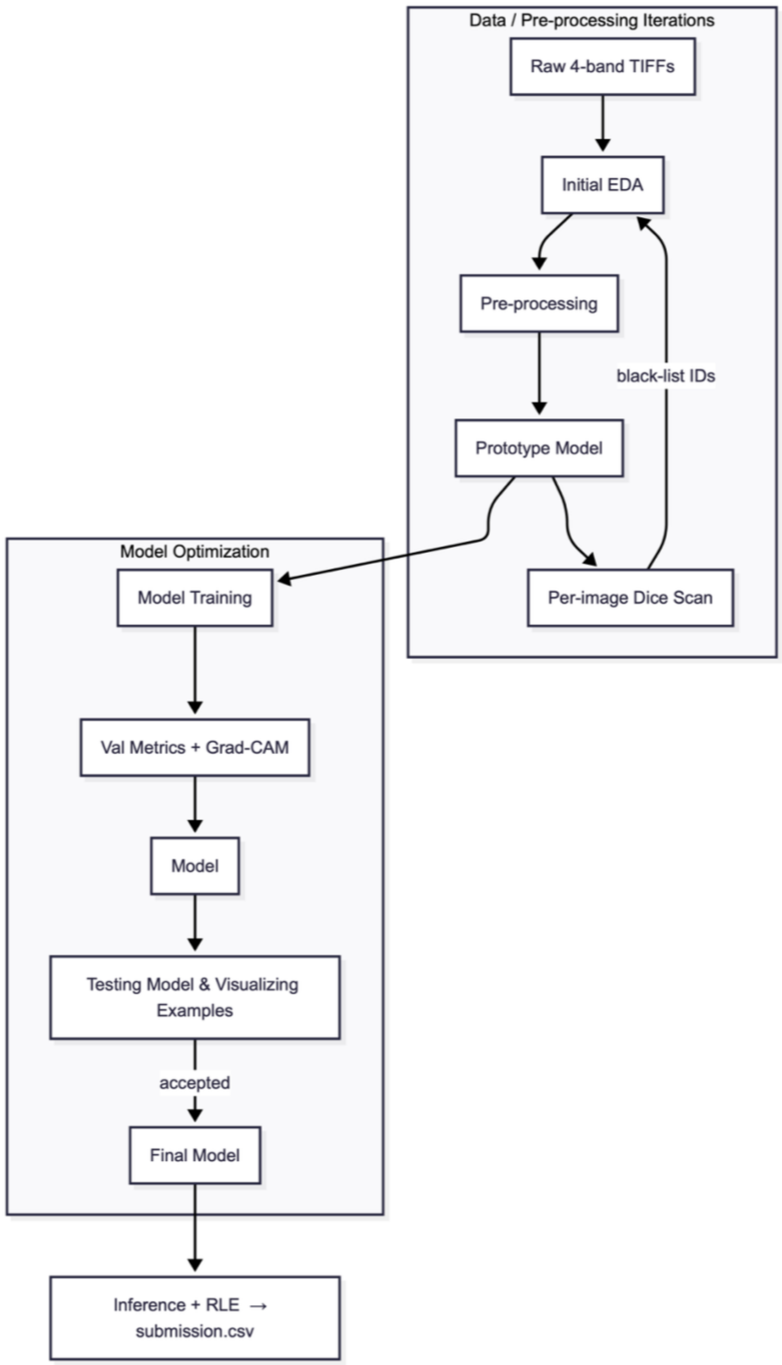
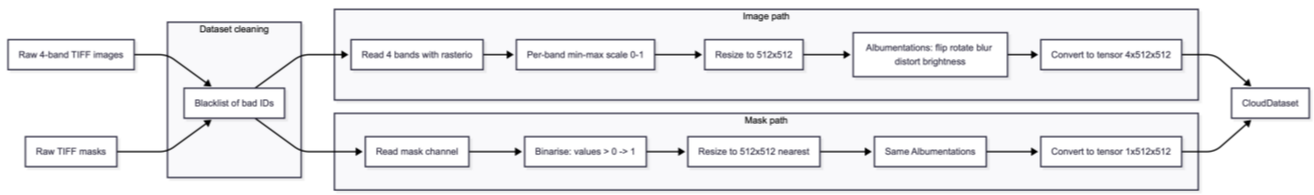
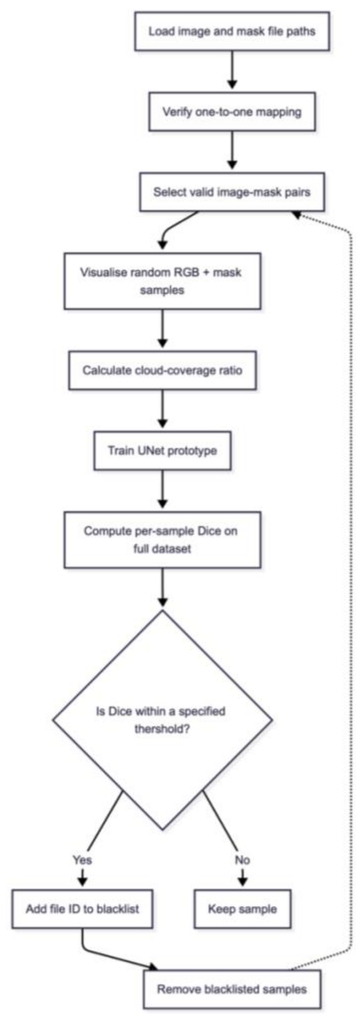
Project Pipeline



Pre-processing & EDA



pre-processing pipeline



EDA pipeline

Stage	Description
Initial EDA	• Counted images/masks, band stats. • Visualized some samples with their masks using different mix of bands. • Counted cloudy, partial clouded, and cloud-free images to see their precentages across the dataset.
Bad-label mining	• Trained an initial U-Net model to filter out bad samples from the dataset based on the dice score. • Manualy revised some dataset samples that have clear masks (cloud-free mask) as we found out some images have been wrongly labeled as cloud-free images.
Augmentation (Albumentations)	• Flip, rotate, blur, grid-distort, brightness/contrast for generalization and avoiding overfitting.

Model Selection & Training Module

We've tried building 3 different models:

- **Random Classifier**.
- **U-Net** offered a strong, efficient baseline and guided most experimentation.
- After hyper-parameter tuning, **DeepLab-v3** surpassed U-Net by +1.9%. Therefore DeepLab-v3 was selected as the final model.

Enhancements & Future Work

- Try alternative encoders.
- Hyper-parameter tuning:
  - systematic sweep of learning-rate schedules, optimizer variants (AdamW, SGD+momentum), weight-decay and loss-term weights.
  - Experiment with different probability thresholds and focal loss γ values.
- K-means exploration:

    - Apply K-means clustering on the 4-band pixel space as an unsupervised prior; results can serve as pseudo-labels or an additional channel.
- Model ensembling / stacking:
    - Combine DeepLab-v3 and U-Net outputs via soft voting or small late-fusion CNN to boost edge consistency.
- Post-processing refinements:
    - Conditional Random Fields (CRF) or Morphological edge-cleaning to remove small, isolated artifacts.

Workload Distribution

Name	ID	Tasks
Hamdy Mohamed Salem	9210365	U-Net Model
Ahmed Ibrahim Hassan	9210020	Classical Machine Learning Model
Ahmed Taha Sayed	9210086	DeepLab-v3 Model